

Detection Report: Custom Rust C2 Beacon

On 7/23/2025 the detection system identified network traffic from a custom Rust C2 Beacon on the compromised machine, talking to a Python Server. This detection leveraged six custom suricata rules and the corresponding Sigma rules. Enriched with MITRE AT&ACK tags. The activity was detected and triaged with Kibana dashboards.

The environment was a Rust C2 Beacon on the compromised debian server. The attacker used a Python server to communicate with the beacon, and used a bash script to curl the commands to the Python server, then over to the beacon. Network traffic was sifted through using six custom suricata rules tailored to the beacon's behavior. Filebeat then sent the files over to elasticsearch, which then populated kibana. Three dashboards were used to evaluate the situation:

- Network Monitoring
- Threat Detection
- MITRE AT&ACK Dashboards.

The attack used HTTP POST requests carrying the payloads. These five tactics were used:

- TA0002 - Execution
- TA0005 - Defense evasion
- TA0007 - Discovery
- TA0011 - Command and Control
- TA0010 - Exfiltration.

Techniques include:

- T1059 - Command and Scripting Interpreter
- T1027 - Obfuscated Files or Information
- T1082 - System Information Discovery
- T1105 - Ingress Tool Transfer
- T1071 - Application Layer Protocol.

Sigma rules aligned with the suricata detections, providing cross-platform alerting and enabling effective SIEM-level monitoring.

Alerts were analyzed using Kibana dashboards that broke down detections by severity and MITRE tactic, facilitating rapid prioritization. The Network Monitoring dashboard revealed consistent outbound connections from the compromised host to the C2 server, matching expected beacon intervals. The MITRE ATT&CK dashboard allowed for granular analysis of attacker techniques, aiding in correlating multiple alerts to a single compromise event.

If left undetected, this beacon activity would allow an adversary persistent command and control access, potentially leading to data exfiltration or further system compromise. To improve detection and response, it is recommended to continue refining Suricata rules, maintain comprehensive MITRE ATT&CK tagging for alert context, and incorporate host-based telemetry to complement network detection.

Figure 1: MITRE ATT&CK Dashboard showing detected tactics and techniques

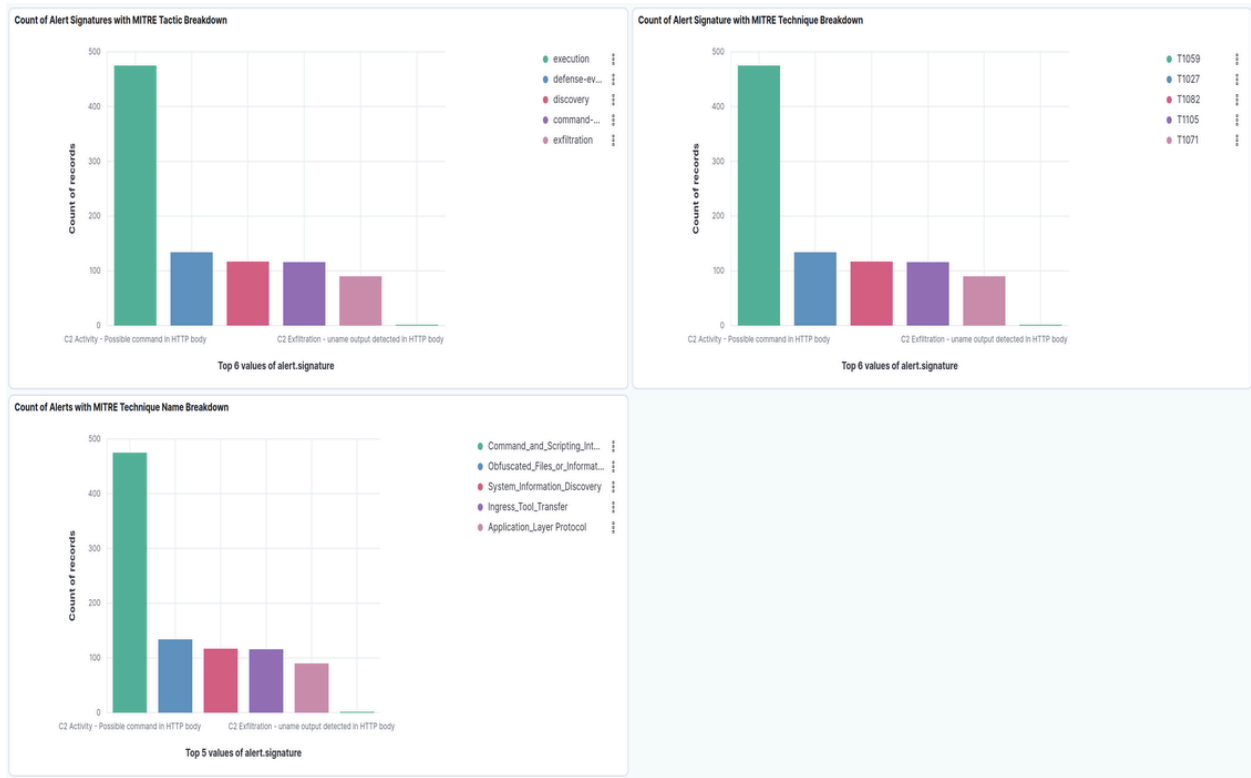
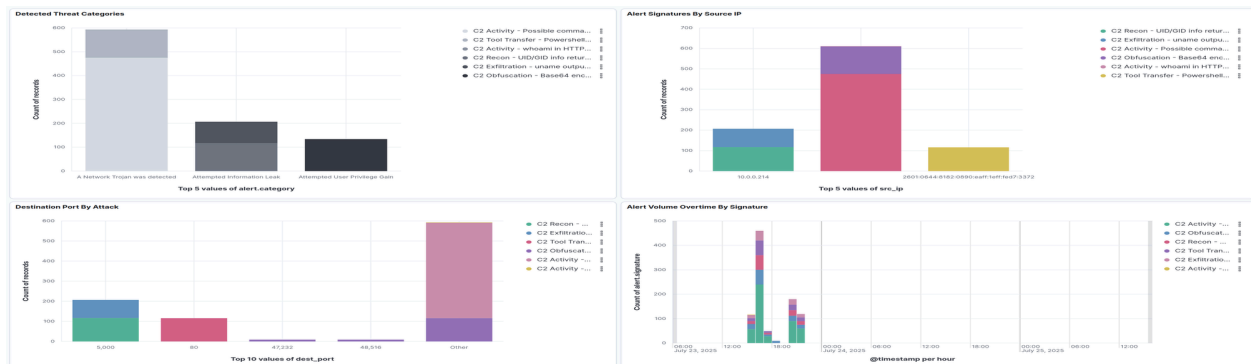


Figure 2,3: Alerts by tactic and severity in Threat Detection dashboard



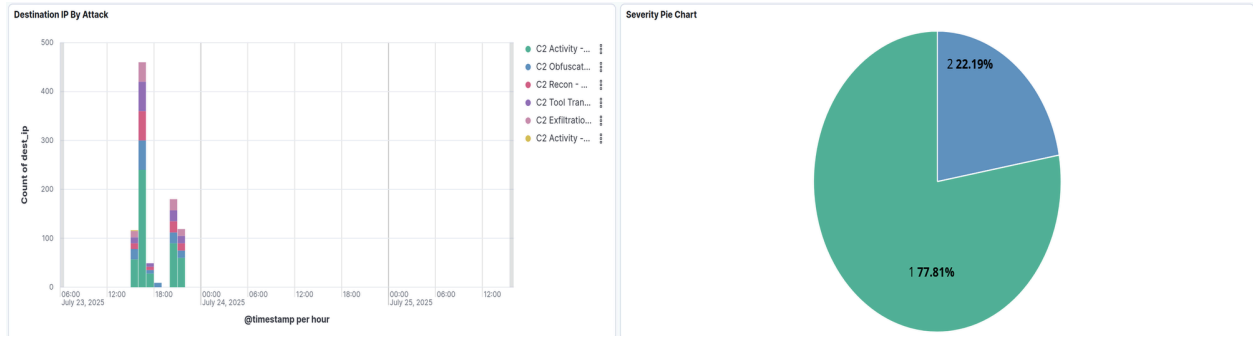


Figure 4,5: Beacon behavior in Network Monitoring dashboard

